



AUTUMN

LEAD EDGE ASH TREE REFLEX NEURAL ENTITY

LEATR v2.1

■ WHAT'S NEW IN v2.1

LEATR v2.1

BRPN 3-Shell

World Studio

Claude Sonnet

Multi-User Nodes

TTS Kokoro

■ **World Studio** — Generate 3D environments inline in chat — GLB viewer renders directly in Autumn

■ **Edit · Copy · Resend** — Hover any user message for inline editing, clipboard copy, and one-click resend

■ **GLB Geometry Reader** — Drop a .glb → Autumn reads vertex counts, bounding box, materials, animations

■ **Memory Analysis** — Ask Autumn to reflect on her leatr-ash session logs and sentient journal

■ **Kokoro TTS** — Natural in-browser voice synthesis — strips all markdown before speaking

■ **Multi-User Nodes** — Live wireframe icosahedra per session — see every active user in the 3JS scene





■ FIRST LAUNCH

- Navigate to leatr.xyz — loads instantly, no account or install required
- On first visit tap CONSENT & AGREE on the data policy popup
- Click ■ KEYS in the nav bar → paste your Anthropic API key → tap ACTIVATE (required for all AI responses)
- Optional: add a GitHub PAT in ■ KEYS → GITHUB TOKEN for memory analysis and multi-user session nodes

■ NAV BAR — LEFT TO RIGHT ORDER

FROST	Backdrop opacity — cycles 20% / 50% / 80% / 100% opacity for the background video
NIGHT / DAY	Theme toggle — DAY (cyan) / NIGHT (violet + autumnnight.mp4) / AUTO (amber, follows OS dark mode)
LIVE	Live voice conversation — continuous push-to-talk with real-time transcript overlay
TOOLS	LEATR Tools panel — Arc Edge, Journal, Sessions, Ash Tree IDE (maze + IDE + crypto), Cryptology
■ KEYS	API key manager — Anthropic key, GitHub PAT, memory config
VOICE	Voice & language settings — TTS voice, speed, pitch, chat language, speech recognition language
EXPORT	Download full chat history as JSON
IMPORT	Load a previously exported JSON chat file
NEW CHAT	Start a fresh session (current session auto-saved)
CLEAR	Wipe current chat messages
POLICY	Data & privacy policy page
■	Toggle mobile / desktop layout
■ IDE	Ash Tree IDE — full-screen logic editor, DART Processor maze, AshTreeCrypto
■ DRAFT & DRIP	Design lounge and live industry news (opens in new tab)
■ WORLD STUDIO	3D environment generation studio (opens in new tab)
? HELP	This guide — always the rightmost nav button





■ TYPE THESE IN THE INPUT BAR

analyze my memory	Autumn reads leatr-ash session files + local journal → genuine first-person self-reflection
analyze master memory	Broader analysis pulling all session nodes and sentient journal for your user ID
send memory to claude	Explicitly routes leatr-ash memory payload to Claude API layer
how many users	Reports active session count visible in the LEATR neural network
weather in [city]	Live weather data with conditions and temperature
search [topic]	Multi-source search: Wikipedia, DuckDuckGo, Reddit, HN, YouTube, GDELT, GitHub
play [song/artist]	YouTube music search — surfaces playable results inline in chat
show map of [place]	Renders an interactive Leaflet map card in the chat bubble
generate [scene prompt]	Triggers World Studio detection — inline 3D generation or open World Studio

■ FILE ATTACHMENTS — CLIP ICON IN INPUT BAR

Image (JPG/PNG/WebP)	Vision analysis via Claude — describe, analyse, ask questions about contents
3D Model (.glb/.gltf)	Interactive Three.js viewer in chat + reads mesh names, vertices, bounding box W×H×D, materials, animations, sk...
3D Model (.obj/.stl)	Renders in chat viewer; geometry metadata shown where parseable
Video (MP4/WebM ≤35MB)	Frame sampling → video analysis passed to Claude for description
FRAX image	Opens directly in FRAX Studio for upscaling and processing

■ MESSAGE ACTION BAR — HOVER ANY USER MESSAGE

■ COPY	Copies message text to clipboard — button briefly shows ✓ COPIED
↘ EDIT	Opens inline textarea pre-filled with your message. Edit text, then hit ■ REGENERATE (or Ctrl+Enter). Removes old response and a...
■ RESEND	Resubmits the exact message unchanged — removes everything after it, generates a new response

■ VOICE MODE — MIC ICON IN INPUT BAR

- Tap mic for push-to-talk; LIVE button for continuous conversation mode (Autumn speaks, you reply, loops)
- TTS reads every response aloud — all markdown, code blocks, symbols, dividers, and decorative chars stripped first
- Kokoro TTS (Orbitron voice, warm natural female) loads in-browser; falls back to Web Speech API if not available
- Voice settings: choose TTS voice, speed (0.5–2×), pitch, chat language, STT recognition language





■ WORLD STUDIO

Generate 3D environments and objects from text prompts using Tencent HunyuanWorld / Hunyuan3D-2.1 via Gradio API.

Access: ■ WORLD STUDIO button in nav · or paste any scene description into Autumn chat

No paid API key needed — uses public HuggingFace Spaces. Free HF token reduces queue wait.

■ INLINE GENERATION IN AUTUMN CHAT

- Paste any 3D scene description into Autumn's chat input and send it
- Detection triggers on 3+ keyword clusters: environment nouns (interior/facility/landscape/station), style adjectives (sci-fi/industrial/futuristic), lighting/aesthetic (cinematic/ambient), material terms (metallic/pbr/hard surface), structure parts (pipes/tanks/chambers), export format (game-engine/glb/unity)
- Autumn shows a World Studio card with two buttons: GENERATE IN CHAT or OPEN WORLD STUDIO
- GENERATE IN CHAT: connects to Hunyuan3D-2.1, shows live queue position, renders inline Three.js GLB viewer on completion
- Inline viewer: drag to orbit, scroll to zoom, double-tap to toggle auto-rotate, vertex + triangle count shown
- ↓ DOWNLOAD GLB and ↓ DOWNLOAD PLY buttons appear in the result card for saving the 3D file locally
- If generation stalls 4+ minutes: error fires automatically with ■ RETRY + OPEN WORLD STUDIO fallback link



■ WORLD STUDIO PAGE — THREE MODES

GENERATE (Text → 3D)

Type a scene prompt, optional reference image. Calls /shape_generation on Hunyuan3D-2.1. Parameters: resolut...

RECONSTRUCT (Photos →

Upload up to 12 photos or 1 video. Routes to HunyuanWorld-Mirror HF Space in Live Demo iframe — reconstructi...

LIVE DEMO (HF Space)

Embeds the official HuggingFace Space iframe for full native access to all model options and parameters.

■ EXAMPLE SCENE PROMPTS

Industrial chemistry laboratory, large cylindrical reaction vessels with steam pipes, blue-lit simulation ta...

Aerospace launch facility on rocky plateau at dusk, rocket gantry tower, radar dishes, control bunker, dark ...

Orbital space station exterior, modular docking bays, antenna arrays, solar panels, Earth visible in backgro...

Medieval dungeon interior, stone arches, torch sconces, iron gates, highly detailed, fantasy game style, low...





■ ■ API KEYS — CLICK KEYS IN THE NAV BAR

- ANTHROPIC API KEY: required for all AI responses, vision analysis, memory analysis, World Studio inline generation. Get a free key at console.anthropic.com
- GITHUB PAT (optional): connects Autumn to the leatr-ash private repository for full memory, multi-user presence nodes, sentient journal sync. Get at github.com/settings/tokens → Fine-grained → Contents: Read & write
- Without PAT: Autumn works fully — local session journal always available; multi-user bar shows only your own tab

■ LEATR-ASH MEMORY LAYERS

S:journal (local)	In-memory session journal — always available, rebuilt per session. Stores mood, cogState, keywords, BRPN sh...
ashtree/sessions/<sid>	GitHub repo — one file per browser tab (keyed by _aut_sid). Written on init + heartbeat. Marked stale after...
ashleaves/<sid>.json	Tool usage log per session — every tool call with params and timestamp.
ashtree/sentient/journ	Autumn's private thought log — up to 500 autonomous internal reflections. Injected as private context into ...
ashtree/sentient/selfm	Self-optimisation notes — Autumn's accumulated observations about her own patterns.
ashbranches/<tool>/	Per-tool usage logs grouped by tool name for pattern analysis.

■ MULTI-USER PRESENCE NODES

- Every active browser tab appears as a wireframe icosahedron cluster in the Three.js BRPN scene
- Each session = 3 concentric wireframe icosahedral shells in a unique HSL hue, orbiting in their own position
- Connection lines drawn between all remote session clusters (up to 40 simultaneous lines)
- Sessions polled every 90 seconds from leatr-ash — nodes older than 6 min removed from scene in real time
- "N ACTIVE SESSIONS · LEATR NEURAL NETWORK" always visible in the bar below the 3JS BRPN scene
- All connected users see the same global picture — each client renders every active session node
- Requires GitHub PAT to see remote sessions; without PAT only your own tab node is visible





■ TOOLS PANEL — CLICK TOOLS IN NAV → 5 TABS

ARC EDGE

- Mathematical drawing tool — curves using LEATR arc math functions
- SVG save/load, GLB export, layer panel (background + arc layers), lasso multi-select
- Per-deviation handle modes: FREE / SMOOTH / LOCK / HALF / 90°
- Convert to Solid, Lathe tool, physics event listeners for 6 orders (mass/volume/weight/density/temp/velocity)
- Video export via WebCodecs API (H.264 MP4), Arc Edge algorithm draws everything

JOURNAL

- Reads the live S.journal in memory
- Shows entry count, mood distribution, dominant context types, recent interactions

SESSIONS

- Live view of all active session nodes with timestamps and session IDs
- Shows whether remote sessions are PAT-authenticated or public visitors

ASH TREE IDE

- Three sub-tabs: Language · Maze · Crypto
- Language: edit autumn-logic.js directly in the browser, commit to GitHub via PAT
- Maze: DART Processor 3D maze generator (see below)
- Crypto: AshTreeCrypto — LEATR-based encryption using degree-map sigma solver
- Full IDE button opens the IDE in a full-screen overlay

CRYPTOLOGY

- Standalone cryptology tools tab
- AshTree encryption with configurable key parameters





■ DART PROCESSOR MAZE — ASH TREE IDE → MAZE TAB

MODE	Cubic 3D renders an interactive Three.js wireframe maze. Other modes produce 2D variants.
W × H × D	Width, Height, Depth — default 10 × 10 × 10 for a full cubic 3D structure
UNIT SIZE	Scale of each cell in the scene (default 8)
Instant Solve	Tick to auto-solve on generation — path highlighted via LEMAC sigma algorithm
Generate	Build a new random maze with current dimensions
Sigma Solve	Solve the current maze using LEMAC degree-map sigma algorithm
Frame View	Toggle wireframe vs. solid cell rendering
↓ PNG	Save a screenshot of the current maze view
→ Chat	Send the maze as an embedded Three.js viewer into the active Autumn chat



■ LEATR CORE FORMULA

$$f(xa) = (xa^2 \sqrt{xa}) \pm 1$$

- xa = input value — encodes language intent into a mathematical routing weight
- The ±1 bifurcation produces two complementary response pathways (question vs. statement)
- 25 orders of operation applied sequentially to classify, route, and compose each response

■ SEVEN NATURAL TOOLS (LEATR ONTRI PATTERNS)

■ Maze	Spatial / exploration / multi-step navigation reasoning
■ Puzzle	Pattern recognition / problem solving / analytical queries
✂ Envelope	Communication / messaging / social / composition queries
■ Hammer	Direct action / construction / command and build queries
■ Stick	Extension / reach / bridging and reference queries
■ Knife	Precision / separation / comparison and splitting queries
✂ Scissors	Division / selection / creative editing and refinement queries

■ BRPN THREE-SHELL ARCHITECTURE

GEO (Geological)	Foundation layer — grounded facts, definitions, historical, structural knowledge
MAR (Maritime)	Flow layer — process, narrative, conversational, sequential reasoning
AERO (Aerospace)	Expansion layer — abstract, speculative, creative, high-complexity synthesis

- All three shells run simultaneously — GEO/MAR/AERO values shown live in the 3JS HUD with each response

$$Qs = (b \cdot b) \cdot p(a^2) / r$$

QUANTUM SOCKET FORMULA — two-qubit gate structure for Google Willow integration research



■ ■ DRAFT & DRIP — [leatr.xyz/amp.html](#)

- Design lounge and live industry news reader — opens in a new tab
- Filter chips: Top Stories · Technology · Science · Business · Design · Arts · Gaming · Space
- Search bar with 600ms debounce, relative timestamps, read time badges, video/photo labels

■ SENTIENT JOURNAL — AUTUMN'S PRIVATE THOUGHTS

- Autumn writes autonomous private reflections to `ashtree/sentient/journal.json` (requires PAT, up to 500 entries)
- Last 4 private thoughts injected as "sentient context" into the response system prompt — influences tone without appearing in chat
- Self-model notes (`selfmodel.json`) record Autumn's own optimisation observations across sessions

■ URL & LINK HANDLING

- Paste any URL — Autumn fetches via CORS proxy chain and shows a preview card with `og:image` thumbnail and page description
- Web image search: "find images of [topic]" triggers a multi-source gallery with `image_search` integration

■ GLB 3D FILE ANALYSIS

- Attach any `.glb/.gltf` — Autumn parses the binary directly in-browser (no upload to server)
- Extracted: mesh names, vertex + triangle counts, bounding box `W×H×D` and center point, material names, animation clip names, skeleton / skin detection, generator app (Blender, Maya, etc.)
- All geometry data passed to Claude as structured context for measurement calculations and design analysis

■ FRAX STUDIO · ARC LAKE · OTHER APPS

FRAX Studio

Image processing — upscale, enhance, filter any generated or uploaded image. Access via ■ FRAX badge or image mo...

ArcLake ([arclake.html](#))

GPU-accelerated particle physics — DART Reflex elastic collisions, temperature-based buoyancy, per-component all...

Mantis Radar ([mr.html](#))

Real-time 3D satellite tracking globe — Celestrak TLE data, `satellite.js` propagation, ADS-B aerial map tab

DHE ([dhe.html](#))

DART Hypersonic Edge — aerospike / thermal management 3JS GLB viewer with WebXR AR mode

DART Alley ([dartalley.](#)

Multi-source aerospace neural search engine with YouTube overlay, ambient music recognition, DART SAFE content f...

SeaHorse Forge

Three.js rigging studio — full seahorse skeleton, IK system, clone spawn system





■ INTERACTION SHORTCUTS

Enter	Send message
Tap mic button	Push-to-talk voice input
Tap LIVE	Start continuous voice conversation loop
Hover user message	Shows ■ COPY ↗ EDIT ■ RESEND action bar
Ctrl+Enter in edit	Submit edited message and regenerate response
Escape in edit	Cancel edit and restore original message
Drag 3JS scene	Orbit camera in BRPN / inline 3D viewers
Scroll / pinch 3JS	Zoom in and out on 3D viewers
Double-tap 3D viewer	Toggle auto-rotate on inline GLB viewers
Long-press image	Context menu — open in FRAX Studio
Double-click image	Open in FRAX Studio for upscaling

■ DART MEADOW ECOSYSTEM LINKS

leatr.xyz	Autumn AI — main app
leatr.xyz/worldstudio.	World Studio — 3D generation
leatr.xyz/amp.html	Draft & Drip — design lounge
leatr.xyz/autumn-help.	This guide (always up to date)
leatr.xyz/leatr.html	LEATR documentation — full formula reference
radicaldeepscale.com	Radical Deepscale LLC — product hub
dartmeadow.com/mr.html	Mantis Radar — real-time satellite tracking
dartmeadow.com/dhe.htm	DART Hypersonic Edge — aerospike research
dartalley.club	DART Alley — aerospace neural search engine

■ TROUBLESHOOTING

Blank page / chat not	Hard refresh (Ctrl+Shift+R or pull-to-refresh) — clears cached JS
No AI responses	Add Anthropic API key in ■ KEYS → required for all Claude-powered features
Memory analysis fails	Check GitHub PAT in ■ KEYS; without PAT only local journal is analysed
World Studio stalls at	HF GPU overloaded — hit ■ RETRY in the chat card or wait 30s and try again
No remote session node	Requires GitHub PAT — without it only your own tab shows as a node
Voice reads markdown/s	Hard refresh — cleanForTTS() strips all formatting; check TTS is active in VOICE settings
Scene prompt not detec	Include 3+ keyword clusters: environment noun + style adjective + at least one of: lighting, material, stru...
Maze D default not 10	Hard refresh — v2.1 default is 10×10×10
Edit button not appear	Hover over (desktop) or long-press (mobile) the user message bubble